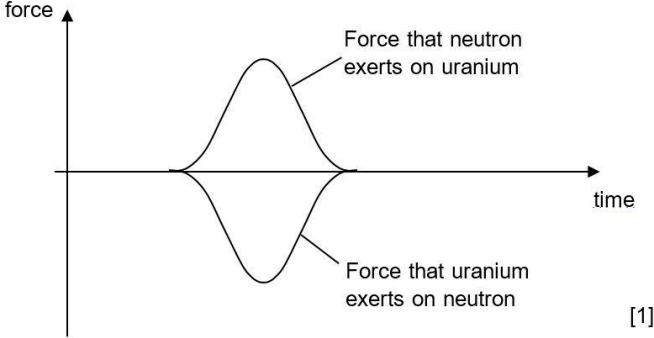


8	(a)	The principle of conservation of momentum states that the total momentum of a system remains constant if no external resultant force acts on the system.		B1
	(b)	(i)		A1
		(ii)	For each curve, the area under the graph is change in momentum of each body. The graphs are mirror images of each other such that the gain in momentum of one is equal to the loss in momentum of the other, resulting in total momentum remaining constant.	B1 B1
	(c)	(i)	1. $Q = 1.2 \text{ MeV}$	A1
			2. $v = \sqrt{E/(1/2 m_e)}$ $= 6.5 \times 10^8 \text{ ms}^{-1}$	A1
			3. $m(\text{Po}) = m(\text{Bi}) - m_e - Q/c^2$ $= [(209.939\text{u}) - (9.11 \times 10^{-31}/1.66 \times 10^{-27}) -$ $(1.2 \times 10^6 \times 1.60 \times 10^{-19}/c^2)]/(1.66 \times 10^{-27} \text{ kg})$ $= 209.937 \text{ u}$ Allow ecf from (c)(i)1. for Q value	C1 C1 A1

		(ii)	Range of values accepted : 0.16 - 0.19 MeV	A1
		(iii)	For a stationary nucleus decaying into the beta particle and daughter nucleus, the conservation of linear momentum requires that $\underline{p_1 = - p_2}$. The sum of kinetic energies will thus be $(p_1)^2/2m_1 + (p_2)^2/2m_2 = E$, which <u>ought to equal the energy released in the reaction</u>, which, if equal to the increase in the total binding energy/decrease in total mass, ought to be constant. The range of beta particle energies and thus the supposed energy released E, seem to suggest that the energy released was not constant, in contradiction to the principle of conservation of energy.	B1 B1
		(iv)	The existence of another undetected particle known as neutrino.	B1
	(d)	(i)	The decay constant refers to the probability of decay of a radioactive nucleus in a time of one second.	B1
		(ii)	Using $A = A_0 e^{-\lambda t}$, $A = 375e^{-\left(\frac{\ln 2}{8.04} \times 2.5 \times 7\right)}$ A = 82.9 Bq	C1 C1 A1
		(iii)	$A = \lambda N$ $A = \left(\frac{\ln 2}{t_{\frac{1}{2}}}\right) \left(\frac{M}{M_m} N_A\right)$ $M = \frac{(375)(8.04 \times 24 \times 60 \times 60)(131 \times 10^{-3})}{(\ln 2)(6.02 \times 10^{23})}$ $\approx 8.2 \times 10^{-17} \text{ kg}$	C1 C1 A1